BMC Plant Biology

RESEARCH

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Organic manure and fertilizer N management strategies improve soil health at different growth stages of pearl millet under pearl millet-wheat sequence

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Abstract

Background Incorporating organic manure improves soil properties and crop productivity. A long-term study started in October 1967 examined the effects of farmyard manure and nitrogen fertilization on the soil at key growth stages of pearl millet in a pearl millet-wheat cropping system over its 51st cycle.

Results Applying 15 Mg of farmyard manure (FYM) per hectare in both growing seasons significantly boosted soil organic carbon (SOC), dissolved organic carbon (DOC), and key nutrients compared to one-season application. SOC levels reached 9.1–11.0% with FYM_{15}, outperforming FYM_{10} and FYM_5. The effectiveness order was $FYM_{15} > FYM_{10} > FYM_5 > FYM_0$. Chemical nitrogen fertilization also enhanced soil properties, with FYM_{15} improving DOC, available nitrogen, and enzyme activities during various growth stages.

Conclusions Principal component analysis (PCA) effectively distinguished soil variables and treatments, with β-glucosidase and arylsulfatase activity (ARA) showing the highest loading (0.294) in the first principal component (PC1), followed by dissolved organic carbon (DOC) at 0.292 and dehydrogenase activity (DHA) at 0.291. A strong polynomial relationship was noted between soil organic carbon (SOC) and various soil properties, with R^2 values between 0.93 and 0.99. The FYM_{15} x N_{120} treatment improved soil health in North-West India's sandy loam soils during both seasons.

Keywords Long-term, Manure, Growth stages, Soil fertility, Enzymes

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Introduction

Pearl millet (*Pennisetum glaucum*)-wheat (*Triticum aestivium*) is a dominant cropping sequence after rice-wheat in India, especially in Haryana's semi-arid regions, covering about 2.26 million ha [1]. While India has achieved food production self-sufficiency, soil quality is declining due to nutrient mining from intensive farming. To address this, revitalizing organic manure alongside fertilizers is essential for maintaining soil fertility and crop productivity [2–4].

Long-term experiments have shown that combining organic manures with fertilizers enhances soil nutrient availability and microbial activity [5–7]. For instance, integrating NPK with FYM has resulted in the highest soil organic carbon content and enzyme activity in various cropping systems [8]. Soil enzymes play a critical role in nutrient transformation and organic residue decomposition, biochemical functions of soils and preserving soil structure, and are sensitive indicators of SOM turnover [9]. The extent of enzyme activity is influenced by soil properties, microbial communities, and environmental factors [10].

Seasonal changes in moisture, temperature and substrate availability affect microbial biomass and enzyme activity, particularly during crop growth stages [11]. Nutrient demand and enzymatic activities fluctuate. More intense microbial activity is often observed at the vigorous crop growth stage, generally the flowering stage, and then gradually declines up to crop harvest [12]. Studies indicate peaks in enzyme activity, such as dehydrogenase and alkaline phosphatase, during specific stages in long-term cropping systems, highlighting the importance of managing nutrients to meet crop demands effectively [13, 14]. Additionally, organic manure application has enhanced soil health indicators over time. In a long-term experiment (> 100 years) of semi-arid sub-tropical conditions in alfisols, organic manure application beneficially favoured SOC accrual, NPK availability, DHA, urease, APA, and ARA over INM and control at various growth stages of maize [15].

The key challenge in developing a sustainable agriculture system is maintaining long-term soil organic matter (SOM), which enhances soil microbial activity and biochemical processes. While inorganic fertilizers boost soil fertility and alter its properties, combining manure and fertilizers can help prevent soil organic carbon (SOC) declines and close the yield gap [16]. Thus, integrating nutrient sources offers a logical approach to sustaining soil fertility and crop productivity [17]. Proper management of nutrient practices and microbiological activities can further enhance productivity. Long-term use of farmyard manure (FYM) and nitrogen fertilizers is expected to improve microbial health and soil fertility in semiarid conditions. Despite various studies examining fertilization's effects on soil health, there is a lack of research on the combined impact of manures and fertilizers on soil properties and enzyme activities [1, 18]. We hypothesized that a combined approach of organic manures and fertilizers would enhance soil fertility and enzymatic activities over time in a pearl millet-wheat system. This investigation aims to assess the long-term effects of levels of FYM during different modes and chemical N application on soil properties (soil reaction, EC, SOC, DOC, available N, P and S) and enzyme activity (DHA, β-glucosidase, Urease, APA and ARA) at various growth stages of pearl millet in Haryana, India.

Materials and methods

Experimental site

The present study was conducted at Research Farm, Department of Soil Science, CCSHAU, Hisar (India), in an ongoing long-term field experiment established in October 1967. The experimental site is situated at a latitude of 29º16'N and longitude of 75º75'E in the North-Western region of India. The study area is characterized by a semi-arid climate with a mean annual precipitation of 443 mm and a mean annual temperature of 24.8ºC. Approximately 90% of rainfall is obtained from the Southwest monsoon during May to October. The lowest mean temperature was recorded in January, while the mean maximum temperature was recorded in May. The soil type was sandy loam, classified explicitly as Typic Ustochrept. The chemical properties of surface soil (0–15 cm) at the start of the experiment (1967) were as follows: soil pH of 8.20, SOC of 0.47%, $CaCO_3$ of 1.10%, available N, P and K were 200.00, 26.00, and 498.00 kg ha^{-1}, respectively [19].

Experimental design

The experiment focuses on the compensation of fertilizer N through FYM under pearl millet-wheat cropping system. The experiment was carried out in a split-plot design with 10 m x 5 m subplots and three replications. The treatments in the main plots involved three methods of applying FYM: applying it to the *Rabi* crop (winter season), to the *Kharif* crop (summer season), and both crops (*Rabi* and *Kharif* seasons). Each crop received three levels of FYM (15, 30, and 45 Mg ha^{-1} yr^{-1} until 2007-08, and 5, 10, and 15 Mg ha^{-1} yr^{-1} from 2008 to 09 onwards, based on dry weight). One FYM control plot was also assigned to each main plot. FYM was added to the plough layer (15 cm) before sowing the crops. The FYM utilized in the experiment was analyzed annually, and the average nutrient composition of FYM applied during 2017-18 was as follows: C (38.10%), N (0.85%), P (0.97%), K (1.87%), S (0.36%), Na (0.68%) and Ca (1.25%). The main plot was subdivided into two subplots, with one receiving no fertilizer N (0 kg N ha^{-1}) and another receiving 120 kg N

ha^{-1} through urea. Application of fertilizer N was divided into two splits: half at the sowing time and the remaining half after 25–30 days after sowing. Throughout the trial, only urea has been used as a chemical fertilizer. Pearl millet (variety HHB 223) was sowed on 5 July 2017, right after the field was prepared following the measures recommended by the institution. The hybrid (HHB 223) used in this experiment was released for cultivation by the Central Variety Release Committee, Ministry of Agriculture Government of India (GOI), and notified vide S.O. 211 (E), dated 29th January 2010. Chaudhary Charan Singh Haryana Agricultural University, Hisar Haryana (India), is the sole proprietor for this hybrid (HHB 223). The crop was irrigated using canal water, and an irrigation of approximately 7.5 cm was applied as and when required based on visual inspection of the field. The crops were harvested around 2–5 cm above ground level, and roots and stubble were incorporated in situ in the field.

Soil sampling

Two sets of soil samples were collected from each plot in triplicate from surface layer (0–15 cm) at the following growth stages of pearl millet: S1 (at sowing), S2 (tillering), S3 (heading), S4 (maturity), and S5 (after harvest). One batch/set of soil samples was dried in the shade by exposure to air, pulverized, filtered through a 2 mm sieve, and stored in a plastic bag under normal environmental conditions for physicochemical analysis. The second fresh, moist soil sample batch/set was placed in a freezer at 4∘C temperature to analyze soil enzyme activity.

Soil fertility and enzyme activities analysis

Soil pH was measured using a glass electrode and potentiometric method (1:2 soil-water suspension). Soil electrical conductivity (EC) was determined using the conductometric method in a 1:2 soil-to-water supernatant [20]. The wet oxidation method was employed to estimate the SOC content [21]. To determine the dissolved organic carbon concentration (DOC), 50 ml distilled water was added to 10 g soil and agitated in a horizontal shaker for one hour, followed by centrifugation for 30 min at 8000 rpm. The DOC was analyzed in filtrate using the dichromate acid oxidation method [22]. The available N was determined using the Kjeldahl-distillation method [23]. Available P was extracted with 0.5 M sodium bicarbonate solution (pH 8.5) and the P content in the extract was then measured at 660 nm wavelength of spectrophotometer [24]. The availability of S was assessed using turbidimetric analysis at 420 nm wavelength [25].

Dehydrogenase activity (DHA) was assessed by estimating the rate at which tri-phenyl formazan (TPF) was produced from 2,3,5-triphenyl tetrazolium chloride (TTC) [26]. The activity of the β-glucosidase enzyme was measured by determining the amount of p-nitrophenol produced when the soil was incubated with p-nitrophenyl-β-D-glucoside (PNG) (pH 6.0) buffer and toluene [27]. The urease activity was measured using a potassium chloride-phenylmercuric acetate (KCl-PMA) solution [28]. The activity of alkaline phosphatase (APA) was assessed using substrate p-nitrophenyl phosphate (PNP) [29]. Aryl sulfatase activity was measured by determining the p-nitrophenol emitted when the soil was incubated with buffered potassium p-nitrophenol sulfate (pH 5.8), acetate buffer, and toluene [30].

Statistical analysis

The statistical analysis was conducted using the software STATISTICA 6.0 by Stat Soft, Inc. (2001). A box plot and principal component analysis (PCA) were used to evaluate the variation accounted for by the principal components (PCs) using Origin (Pro) Version 2024. The two most prominent principal components, representing the most significant variation, were visually illustrated in a two-dimensional graph.

Results

Soil reaction

Soil pH ranged from 7.49 to 7.77, 7.38 to 7.71, 7.24 to 7.68, 7.42 to 7.60, and 7.38 to 7.73 at S1, S2, S3, S4, and S5 of pearl millet growing period, respectively (Table 1). Modes of FYM application showed a significant impact on soil pH regardless of the growth stages of pearl millet. Among modes of FYM application, the significantly lowest soil pH was recorded with FYM application during both seasons at all the stages, and the order was obtained: *rabi* > *kharif* > both seasons. Levels of FYM application exhibited a significant influence on soil pH. The highest reduction of soil pH was recorded with the addition of 15 Mg FYM ha^{-1} compared to its lower application rates at all studied stages of pearl millet (Table 1). Mode x level of FYM significantly interacted with soil reaction during the growing period of pearl millet (Supplementary Table 1). Soil pH ranged from 7.41 to 7.77, 7.30 to 7.71, and 7.10 to 7.68 at stages 1, 2 and 3, respectively. Regardless of stages, the lowest soil pH among different treatment combinations was observed with both seasons x FYM_{15}. Further, the FYM x fertilizer N level also interacted significantly with soil pH, ranging between 7.47 and 7.78 (Supplementary Fig. S1). All three factors showed decreased soil pH at S5 over S1 (Table 1). A box plot showed 50% soil pH values ranging between 7.76 and 7.48 at S1 as opposed to 7.66 and 7.41 at S5. However, the lowest pH was found at S3, with 50% values ranging from 7.63 to 7.30 (Fig. 1A).

Soil electrical conductivity

Soil EC ranged from 0.29 to 0.46, 0.31 to 0.55, 0.33 to 0.58, 0.27 to 0.43, and 0.26 to 0.39dS m^{-1} at S1, S2, S3,

Table 1 Long-term impact of mode, level of FYM, and fertilizer N application on soil reaction, electrical conductivity (EC), and organic carbon at various growth stages of pearl millet (the subscript figures in treatment indicate the dose of FYM in t ha^{-1} and the dose of fertilizer nitrogen in kg ha^{-1})

Treatment	Season/level of input	Soil pH (1:2)					Soil EC (dS m^{-1})					Soil organic carbon (%)				
		Pearl millet growth stage														
		S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
Mode of FYM application	Rabi	7.71	7.56	7.51	7.60	7.56	0.39	0.42	0.45	0.32	0.28	1.26	1.32	1.37	1.30	1.23
	Kharif	7.60	7.55	7.47	7.58	7.54	0.34	0.44	0.47	0.34	0.31	1.21	1.34	1.40	1.32	1.25
	Rabi and Kharif, both seasons	7.53	7.44	7.33	7.52	7.48	0.41	0.46	0.49	0.37	0.34	1.40	1.49	1.53	1.46	1.43
CD (p = 0.05)		0.04	0.04	0.04	0.05	0.06	0.02	0.03	0.03	0.02	0.03	0.06	0.06	0.06	0.05	0.04
Level of FYM (t ha^{-1})	FYM_0	7.77	7.71	7.68	7.55	7.73	0.29	0.31	0.33	0.27	0.26	0.54	0.57	0.59	0.55	0.52
	FYM_5	7.62	7.53	7.48	7.59	7.53	0.36	0.41	0.44	0.31	0.27	1.40	1.52	1.58	1.49	1.42
	FYM_{10}	7.57	7.45	7.35	7.51	7.47	0.42	0.49	0.52	0.36	0.32	1.53	1.65	1.70	1.63	1.55
	FYM_{15}	7.49	7.38	7.24	7.42	7.38	0.46	0.55	0.58	0.43	0.39	1.69	1.80	1.87	1.78	1.72
CD (p = 0.05)		0.05	0.04	0.05	0.05	0.07	0.02	0.03	0.03	0.03	0.04	0.07	0.06	0.07	0.06	0.05
Level of fertilizer N (kg ha^{-1})	N_0	7.62	7.53	7.46	7.58	7.54	0.37	0.43	0.46	0.33	0.30	1.24	1.33	1.38	1.31	1.25
	N_{120}	7.60	7.50	7.42	7.55	7.51	0.39	0.46	0.49	0.36	0.32	1.33	1.43	1.49	1.41	1.35
CD (p = 0.05)		*NS*	0.03	0.04	*NS*	*NS*	0.01	0.02	0.02	0.02	*NS*	0.06	0.05	0.05	0.03	0.02

S4 and S5, respectively (Table 1). Modes, levels of FYM, and chemical N application showed a significant impact on soil EC. In both seasons, the FYM application showed the highest value of EC (0.34 to 0.49dS m^{-1}) and significantly increased EC compared to the *rabi* season. Over S1, soil EC decreased by 28.2, 8.8, and 17.1% with FYM application during *rabi*, *kharif*, and both seasons. The FYM application rates significantly impacted soil EC at all stages, and treatment FYM_{15} showed the highest soil EC, significantly higher (9.5 to 21.9%) than FYM_{10} at various growth stages. Fertilizer N application increased considerably soil EC (except S5) by 5.4 to 9.1% at different stages. The highest soil EC (0.52dS m^{-1}) was found when 15 Mg FYM ha^{-1} was applied during both seasons, followed by the same dose of FYM given in *rabi* (0.48dS m^{-1}) and *kharif* (0.40dS m^{-1}) season (Supplementary Fig. S2). A box plot showed that 50% values of soil EC ranged between 0.44 and 0.30 dS m^{-1} at S1 as opposed to 0.35 to 0.26 dS m^{-1} at S5 (Fig. 1B).

Soil organic carbon

During the 51st cycles of pearl millet under the pearl millet-wheat sequence, SOC ranged between 0.54 and 1.60, 0.57 to 1.80, 0.59 to 1.87, 0.55 to 1.78 and 0.52 to 1.72% at S1, S2, S3, S4 and S5, respectively (Table 1). Modes of FYM, levels of FYM, and fertilizer N application significantly influenced SOC content in these sandy loam soils. Irrespective of the stages of pearl millet, both seasons of FYM application showed the highest SOC and obtained order, both seasons > *rabi* season > *kharif* season. Application of 15 Mg FYM ha^{-1} showed significantly the highest SOC content (S: 10.5, S2: 9.1, S3: 10.0, S4: 9.2, and S5: 11.0%) over 10 t FYM ha^{-1} application. Long-term data indicated that modes x levels of FYM significantly influenced SOC content at stages 1, 4, and 5 of the pearl millet growing period (Supplementary Table 1). Soil OC ranged from 0.54 to 1.86, 0.55 to 1.92, and 0.52 to 1.91% at S1, S4 and S5, respectively. Both seasons x FYM_{15} exhibited significantly higher SOC content at these growth stages than single season x FYM_{15}. Soil OC slightly increased with the growing period up to S3 and decreased thereafter; however, there was a slight increase in SOC at S5 over S1. A box plot indicated that 50% values of SOC ranged from 1.61 to 0.95% at S1 (Fig. 1C).

Soil-dissolved organic carbon

Dissolved OC in soil ranged from 50.5 to 243.1, 66.1 to 282.5, 69.0 to 298.5, 57.5 to 277.3, and 45.6 to 268.6 mg kg^{-1} at different growth stages of pearl millet (Supplementary Table 2). The application of FYM in both seasons significantly increased DOC content compared to the single-season application, regardless of the stages. Application of 15 Mg FYM ha^{-1}significantly accrued DOC content (S1: 33.4, S2: 25.0, S3: 21.4, S4: 44.0 and S5:

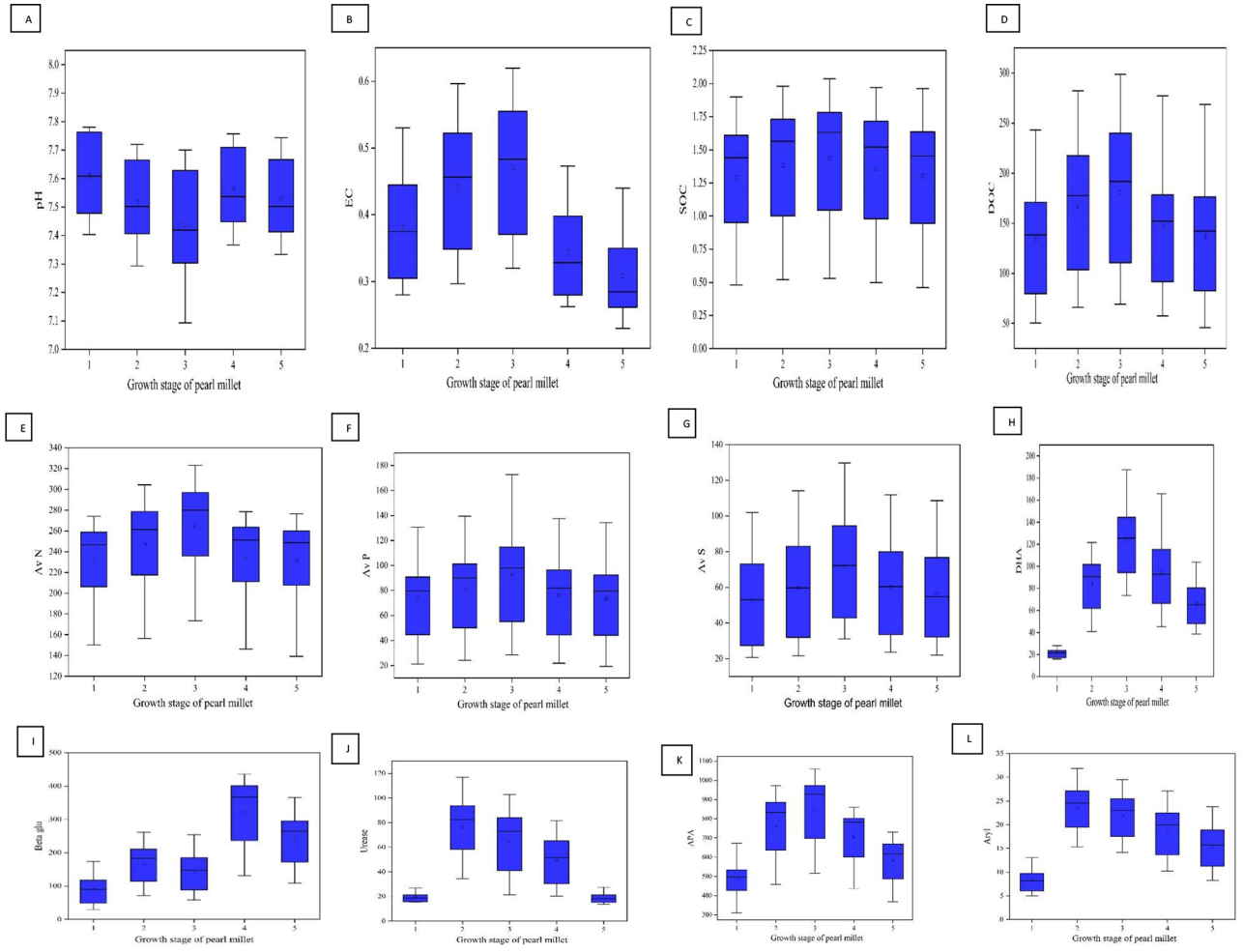


Fig. 1 Box plot of soil pH (**A**), EC (**B**), SOC (**C**), DOC (**D**), available N (**E**), P (**F**), S (**G**), dehyrdogenase activity (**H**), β-Glucosidase (**I**), urease (**J**), alkaline phosphatase activity (**K**) and aryl sulfatase activity (**L**) at different growth stages of pearl millet in soils under pearl millet-wheat cropping system

50.3%), over application of 10 Mg FYM ha^{-1}. Similar to SOC, adding chemical N significantly increased the DOC content by 14.4, 11.4, 10.9, 10.8, and 12.8% at different growing stages of pearl millet. Interaction of mode x level of FYM varied from 57.7 to 235.1, 72.7 to 276.5, 74.6 to 294.1, 64.1 to 265.2, and 56.7 to 256.1 mg kg^{-1} at stages 1–5, respectively (Supplementary Table 2). Regardless of growth stages, the application of 15 Mg FYM ha^{-1} during both seasons was significantly superior among all other treatment combinations. Interaction of both seasons x N_{120} was significantly the best combination for increasing DOC in soil (Supplementary Table 2). At stages 2 and 3, DOC ranged from 66.1 to 244.6 and 69.0 to 269.3 mg $kg^{-1,}$ and FYM_{15} x N_{120} was significantly superior for soil DOC content (Supplementary Table 2). The abundance of DOC with nutrient management practices was lowest at S1 and highest at S3; a box plot exhibited that 50% of DOC ranged between 239.9 and 110.7% at S3 (Fig. 1D).

Available nutrients in the soil

Available nitrogen

Among different treatments, soil available N ranged between171 to 259,179 to 278, 196 to 296, 169to 264, and 164to 261 $kgha^{-1}$atS1, S2, S3, S4 and S5, respectively (Table 2). Modes of FYM application exhibited a significant impact on available N. In both seasons, FYM application enhanced the N availability of soil by 4.5 to 10.0% over the *rabi* season at different growth stages. The application of 15 Mg FYM ha^{-1} accounted for the highest available N, but it could not significantly sustain N availability over the application of 10 Mg FYM ha^{-1}. Chemical N fertilization (N_{120}) proved best to increase N availability (10.2 to 11.5%) in soil at different growing stages of pearl millet in these semi-arid soils of Haryana. Treatment both seasons x FYM_{15} was found considerably superior to enhance available N (316 kg ha^{-1}) in soil as compared to the same dose of FYM applied in either single season (Supplementary Fig. S3). FYM x fertilizer N level significantly interacted with available N of soil,

Table 2 Long-term impact of mode, level of FYM, and fertilizer N application on available nutrients (NPS, kg ha^{-1}) at various growth stages of pearl millet (the subscript figures in treatment indicate the dose of FYM in t ha^{-1} and the dose of fertilizer nitrogen in kg ha^{-1})

Treatment	Season/level of input	Available *N* (kg ha^{-1})					Available *P* (kg ha^{-1})					Available *S* (kg ha^{-1})				
		Pearl millet growth stage														
		S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
Mode of FYM application	Rabi	229	238	254	229	226	66.2	71.8	78.4	65.5	63.6	50.5	54.1	64.5	54.1	51.0
	Kharif	226	243	261	231	227	63.8	74.4	83.2	68.4	64.8	47.8	57.2	69.6	57.1	54.1
	Rabi and Kharif, both seasons	240	259	279	243	239	90.1	97.1	115.8	94.2	91.9	60.4	68.2	81.3	69.1	65.1
CD (p = 0.05)		9	8	8	9	11	3.2	3.9	4.9	4.6	3.9	3.3	3.4	4.5	3.2	3.7
Level of FYM (t ha^{-1})	FYM_0	171	179	196	169	164	23.7	26.9	31.4	24.7	21.4	21.9	23.4	35.0	26.9	25.4
	FYM_5	243	260	279	248	245	76.8	85.3	94.6	77.8	77.5	42.0	49.5	60.3	49.2	46.1
	FYM_{10}	253	270	288	256	254	91.9	101.3	114.6	95.0	92.6	63.7	71.6	85.2	73.0	67.4
	FYM_{15}	259	278	296	264	261	101.2	111.0	129.3	106.7	102.3	84.1	94.8	106.7	91.3	88.1
CD (p = 0.05)		10	9	9	10	13	3.7	4.5	5.6	5.3	4.4	3.9	3.9	5.2	3.8	4.3
Level of fertilizer N (kg ha^{-1})	N_0	220	233	252	222	218	70.7	78.1	88.6	72.8	70.8	48.5	55.1	65.9	54.9	51.5
	N_{120}	243	260	278	246	243	76.1	84.2	96.4	79.3	76.1	57.4	64.6	77.6	65.3	62.0
CD (p = 0.05)		5	7	7	5	10	2.2	2.3	2.8	3.0	2.6	3.2	2.8	3.3	2.9	3.3

ranging from 150 to 266, 156 to 290, 174 to 306, and 146 to 270 kg ha^{-1} at stages 1 to 4, respectively (Supplementary Fig. S4). The addition of 15 Mg FYM and 120 kg fertilizer N ha^{-1} showed the highest available N in the soil. Available N was highest at S3 of the pearl millet growing period, and a box plot exhibited that 50% values of available N ranged from 297 to 236 kg ha^{-1} at S3 (Fig. 1E).

Available phosphorus

Available Pin soil ranged from 23.7 to 101.2, 26.9 to 111.0, 31.4 to 129.3, 24.7 to 106.7, and 21.4 to 102.3 kg ha^{-1}at pearl millet growing S1, S2, S3, S4 and S5, respectively (Table 2). All three factors exhibited significant effects on P availability in soils. Irrespective of the growing stages, the application of FYM during both seasons significantly increased available P by 35.2 to 47.7% over the *rabi* season at various growth stages of pearl millet (Table 2). However, the FYM application in *kharif* was slightly better than the *rabi* season application, except for S1. During the 51st cycle of pearl millet under pearl millet-wheat cropping, FYM applications remarkably influenced P availability in soil and followed the order, $FYM_{15} > FYM_{10} > FYM_5 > FYM_0$ (Table 2). Relative to applying 10 Mg FYM ha^{-1}, 15 Mg FYM ha^{-1} application significantly enhanced available P by 10.1, 9.6, 12.8, 12.3, and 10.5% at S1, S2, S3, S4, and S5, respectively. Further, chemical N application increased available P by 7.6, 7.8, 8.8, 8.9 and 7.5% at pearl millet growing S1, S2, S3, S4, and S5, respectively. Interaction of mode x level of FYM for soil available P varied from 23.7 to 127.2, 26.9 to 135.4, 31.4 to 169.0, 24.7 to 134.6, and 21.4 to 130.9 kg ha^{-1} at stage 1–5, respectively (Supplementary Table 1). Treatment in both seasons x FYM_{15} was significantly superior to other combinations for increasing soil P availability throughout the growing period of pearl millet. A box plot also indicated that available P was lowest at S1 and highest at S3, whereas 50% ranged from 114.8 to 55.21 kg ha^{-1} at S3 (Fig. 1F).

Available sulphur

Available S in soil varied between21.9 to 84.1, 23.4 to 94.8, 35.0 to 106.7, 26.9 to 91.3, and 25.4 to 88.1 kg ha^{-1} at pearl millet growing S1, S2, S3, S4 and S5, respectively (Table 2). Regardless of stages, applying FYM during both seasons significantly enhanced available S content in soil (19.6 to 27.7%) compared to *rabi* season and obtained the order, both seasons > *kharif* season > *rabi* season. Over the application of 10 Mg FYM ha^{-1}, its higher rate (15MgFYM ha^{-1}) application significantly increased available S by 32.0, 32.4, 25.2, 25.1 and 30.7% at S1, S2, S3, S4, and S5, respectively. Among various stages of pearl millet, chemical N fertilized plots showed a significant increase (17.2 to 20.4%) in soil available S than unfertilized (N_0) plots. Interaction of mode x level of

FYM significantly impacted available S, and it ranged between 21.9 and 97.3, 23.4 to 108.6, 35.0 to 123.3, 26.9 to 107.7, and 25.4 to 103.2 kg from stages 1–5, respectively (Supplementary Table 1). Irrespective of growth stages, both seasons x FYM_{15} significantly performed better than the rest of the treatment combinations in increasing soil S availability (Supplementary Table 1). The periodic abundance of available S was similar to that observed for available P, and a box plot revealed that 50% of values of available S ranged from 94.6to42.9 at S3 (Fig. 1G).

Soil enzyme activities

Dehydrogenase activity

The impact of different modes, levels of FYM, and fertilizer N application on dehydrogenase activity (DHA) in soil ranged from 16.1 to 24.1, 48.3 to 108.9, 80.1 to 154.8, 52.5 to 126.8 and 40.8 to 83.3 µg TPF g^{-1} 24 h^{-1} at pearl millet growing S1, S2, S3, S4 and S5, respectively (Table 3). Various modes of FYM application significantly improved DHA activity in the soil and followed the order: both seasons > *kharif* season > *rabi* season (except S1). Both seasons' FYM application significantly improved DHA by 14.4 to 37.1% over the *rabi* season FYM application. Treatment FYM_{15} enhanced the DHA by 12.4, 11.1, 16.1, and 12.4% at S2, S3, S4, and S5 over treatment FYM_{10}. Regardless of growing stages, chemical N application was found essential and significantly helped to improve DHA at different stages by 10.6 to 14.7%. Therefore, all three factors individually played crucial roles in enhancing and maintaining DHA during the pearl millet growing period (Table 3). Modes x level of FYM significantly interacted with DHA activity at stages 4 and 5, and DHA ranged from 80.1 to 176.6, 52.5 to 157.6, and 40.8 to 99.9 µg TPF g^{-1} 24 h^{-1}, respectively (Supplementary Table 3). The impact of 15MgFYM application during both seasons significantly improved DHA at these stages compared to other combinations (Supplementary Table 3). The trend of DHA was similar to that recorded for SOC and available nutrients, and a box plot showed that 50% of DHA values ranged between 144.4 and 94.3 µg TPF g^{-1} 24 h^{-1} at S3(Fig. 1H).

β-glucosidase activity

The β-glucosidase activity of soil varied from 35.7 to 134.1, 80.4to 224.3, 64.3 to 200.6, 146.7 to 410.2, and 118.2 to 312.9 µg PNP g^{-1} h^{-1}at pearl millet growing S1, S2, S3, S4 and S5, respectively (Table 3). Irrespective of the stages, a significant effect of modes, levels of FYM, and chemical N (except S5) application was recorded on β-glucosidase activity under the pearl millet growing period. Regardless of stages, both seasons FYM application showed higherβ-glucosidase activity (S1: 36.5; S2: 22.4; S3: 44.1; S4: 10.8 and S5: 19.4%) over *rabi* season and seasons obtained order, both seasons > *kharif* > *rabi*.

Table 3 Long-term impact of mode, level of FYM and fertilizer N application on soil enzymes activity at different growth stages of pearl millet (The subscript figures in treatment indicate the dose of FYM in t ha^{-1} and the dose of fertilizer nitrogen in kg ha^{-1})

Treatment	Season/level of input		Dehydrogenase (DHA,µg TPF g^{-1} 24 h^{-1})					β-Glucosidase activity (µg PNP g^{-1} h^{-1})					Urease activity (µg NH_4^+-N g^{-1} h^{-1})				
			Pearl millet growth stage														
			S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
Mode of FYM application	Rabi		20.1	77.9	111.9	80.3	56.9	81.0	154.4	122.8	306.5	221.0	18.4	71.4	57.6	44.1	16.8
	Kharif		19.5	83.6	121.8	92.9	63.2	74.5	160.5	132.0	320.0	235.5	17.6	74.9	64.5	48.9	18.2
	Rabi and Kharif, both seasons		23.0	91.3	137.2	107.6	78.0	110.6	189.0	177.0	339.7	263.8	20.8	83.7	71.7	55.5	21.0
CD (p = 0.05)			2.4	8.1	8.1	7.8	6.3	6.8	6.0	14.4	11.8	19.0	1.8	3.4	3.5	2.7	2.2
Level of FYM (t ha^{-1})	FYM_0		16.1	48.3	80.1	52.5	40.8	35.7	80.4	64.3	146.7	118.2	15.5	40.6	23.5	21.6	14.2
	FYM_5		20.9	82.9	120.4	85.8	65.9	80.0	166.7	142.4	346.8	248.5	18.2	82.1	71.8	49.9	18.4
	FYM_{10}		22.3	96.9	139.3	109.2	74.1	105.0	200.3	168.3	384.6	280.9	20.2	88.7	78.1	59.1	19.9
	FYM_{15}		24.1	108.9	154.8	126.8	83.3	134.1	224.3	200.6	410.2	312.9	21.8	95.2	84.9	67.4	22.2
CD (p = 0.05)			2.8	9.3	9.4	9.0	7.3	7.8	7.0	16.6	13.7	22.0	2.1	3.9	4.1	3.2	2.5
Level of fertilizer N (kg ha^{-1})	N_0		19.8	78.9	115.2	87.4	61.9	81.6	160.0	137.0	311.3	230.9	17.6	70.0	58.6	43.7	17.5
	N_{120}		21.9	89.5	132.1	99.8	70.2	95.8	175.9	150.8	332.8	249.3	20.2	83.4	70.6	55.3	19.9
CD (p = 0.05)			1.8	6.3	6.5	6.1	5.0	6.0	5.4	13.2	9.7	NS	1.8	3.1	2.9	2.8	2.1

Application of 15 Mg FYM ha^{-1} was found to be significantly better than other lower doses of FYM and increased β-glucosidase activity by 27.7, 12.0, 19.2, 6.7, and 11.4% at stages 1 to 5, respectively, over 10 Mg FYM ha^{-1}. Treatment N_{120}significantly increasedβ-glucosidase activity by 17.4, 9.9, 10.1, 6.9, and 8.0% at stages 1–5, respectively, over treatment N_0. The interaction effect of modes x FYM on β-glucosidase activity ranged from 35.7 to 167.5, 80.4 to 253.3, and 64.3 to 246.0(μg PNP g^{-1} h^{-1}) at S1, S2, and S3, respectively (Supplementary Table 3). The impact of both seasons x FYM_{15} on β-glucosidase activity was significantly higher than their other combinations. Relative abundance showed an irregular trend, and a box plot exhibited thatβ-glucosidase activity was highest at S4, and a box plot showed that 50% values of β-glucosidase activity ranged from 400.9 to 236.6 at S4 (Fig. 1I).

Urease activity

Urease activity of soil fluctuated between 15.5 and 21.8, 40.6 to 95.2, 23.5to 84.9, 21.6 to 67.4, and 14.2 to 22.2 μg NH_4^+-N g^{-1} h^{-1} at pearl millet growing S1, S2, S3, S4 and S5, respectively (Table 3). Relative to either single season, both seasons' FYM applications significantly improved urease activity and application of FYM in both seasons increased urease activity by 13.0, 17.2, 24.5, 25.9, and 25.0% at stages 1 to 5, respectively, over the *rabi* season. Overtreatment, $FYM_{10,}$ and treatment FYM_{15} significantly increased urease activity by 7.9, 7.3, 8.7, 14.0, and 11.6% at stages 1–5, respectively. Similar to available N, plots treated with chemical N showed a significant increase in urease activity by 14.8, 19.1, 20.5, 26.5, and 13.7% at stages 1–5, respectively, compared to control plots (N_0). The interaction of mode x level of FYM was significant at stages 2, 3, and 4, and it ranged between 40.6 and 107.1, 23.5 to 93.6, and 21.6 to 75.9 μg NH_4^+-N g^{-1} h^{-1}, respectively (Supplementary Table 3). Further, the level of FYM x fertilizer N also considerably interacted with urease activity, and it varied between 21.2 and 92.8 μg NH_4^+-N g^{-1} h^{-1} and 20.1 to 74.5 μg NH_4^+-N g^{-1} h^{-1} at stages 3 and 4, respectively (Supplementary Fig. S5). Treatment FYM_{15} x N_{120} was best for increasing urease activity at stages 3 and 4 compared to the rest of the treatment combinations. Among the stages of pearl millet growth, urease activity followed the order S2 > S3 > S4 > S5 > S1, and a box plot showed that 50% values of urease activity ranged from 93.8 to 58.3 μg NH_4^+-N g^{-1} h^{-1} at S2 (Fig. 1J).

Alkaline phosphatase activity

Among different treatments, APA ranged between354to 563, 472 to 911, 530 to 1005, 449 to 820, and 396 to 691 μg PNP g^{-1} h^{-1}at pearl millet growing S1, S2, S3, S4, and S5, respectively (Table 4). Among different stages, APA increased by 16.4, 5.7, 6.0, 4.6, and 8.1% with both seasons FYM application over *rabi* season and obtained order, both seasons > *kharif* > *rabi* (except S1). Regardless of the stages, each level of FYM showed a favourable impact on APA, and 15 Mg FYM ha^{-1} application was superior to enhancing APA than its lower doses. Chemical N application also significantly increased APA at different stages of pearl millet (Table 4). Among interactive effects, mode x level of FYM played a beneficial role on APA and varied between 354 and 653, 472 to 949, 530 to 1048, 449 to 851, and 396 to 719 μg PNP g^{-1} h^{-1} at stages1 to 5, respectively (Supplementary Table 3). Further, FYM x fertilizer N also significantly interacted at stages 1 and 5, and it ranged from 311 to 576 and 368 to 704 at these stages, respectively (Supplementary Fig. S6). Treatment FYM_{15} x N_{120} was significantly superior for

Table 4 Long-term impact of mode, level of FYM, and fertilizer N application on soil enzymes (alkaline phosphatase activity, APA and aryl sulfatase activity, ARA) activity at different growth stages of pearl millet (the subscript figures in treatment indicate the dose of FYM in t ha^{-1} and the dose of fertilizer nitrogen in kg ha^{-1})

Treatment	Season/level of input	Alkaline phosphatase activity (μg PNP g^{-1} h^{-1})					Aryl sulfatase activity (μg PNP g^{-1} h^{-1})				
		Pearl millet growth stage									
		S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
Mode of FYM application	Rabi	464	746	823	694	558	7.80	22.16	20.67	17.06	14.23
	Kharif	456	761	847	702	575	7.28	23.30	21.78	18.66	14.97
	Rabi and Kharif, both seasons	540	789	872	726	603	9.93	25.29	23.27	20.16	16.96
CD (p = 0.05)		7	9	5	7	7	1.40	1.46	1.29	1.52	1.11
Level of FYM (t ha^{-1})	FYM_0	354	472	530	449	396	5.32	16.33	14.69	10.67	9.17
	FYM_5	497	817	905	768	591	7.69	23.96	22.07	18.61	14.47
	FYM_{10}	533	862	949	792	637	9.08	25.62	24.31	21.02	17.11
	FYM_{15}	563	911	1005	820	691	11.26	28.43	26.57	24.22	20.79
CD (p = 0.05)		9	10	6	8	8	1.62	1.68	1.48	1.75	1.28
Level of fertilizer N (kg ha^{-1})	N_0	469	753	833	699	562	7.92	22.82	21.30	18.02	14.64
	N_{120}	505	778	862	715	595	8.76	24.35	22.52	19.23	16.13
CD (p = 0.05)		6	7	7	5	5	0.51	0.58	0.84	0.91	0.84

APA in soil than other combinations involving the use of FYM-coupled fertilizer N under pearl millet (Supplementary Fig. S6). APA increased up toS3 and declined with advancement of the growth stage. A box plot showed that 50% values varied from 973 to 698 µg PNP g^{-1} h^{-1} at growth S3 of pearl millet (Fig. 1K).

Aryl sulfatase activity

Among different treatments, ARA ranged between 5.32 and 11.26, 16.33 to 28.43, 14.69 to 26.57, 10.67 to 24.22, and 9.17 to 20.79 µg PNP g^{-1} h^{-1} at pearl millet growing S1, S2, S3, S4 and S5, respectively (Table 4). Similar to soil available S, modes of FYM application significantly influenced the ARA and obtained the order, both seasons > *kharif* > *rabi* (except S1). Among different growth stages, ARA increased by 27.3, 14.1, 12.6, 18.2, and 19.2% with both seasons FYM application over *rabi* season. The addition of 15 Mg FYM ha^{-1} showed an increase of ARA (9.3 to 24.0%) compared to 10 Mg FYM ha^{-1} application at different stages. Chemical N application also significantly increased ARA from 5.7 to 10.6% at various stages of pearl millet. It was lowest at S1 and sharply increased at S2; however, thereafter, a regular decrease was obtained up to S5, and a box plot showed that 50% values varied from 27.1 to 19.5 µg PNP g^{-1} h^{-1} at growth S2 of pearl millet (Fig. 1L).

Discussion

Soil reaction and electrical conductivity

Long-term nutrient management practices decreased soil pH under the pearl millet-wheat sequence compared to the antecedent value due to the release of organic acids during the decomposition of organic matter. The decrease in soil pH with the addition of FYM might also be due to the oxidation of organic matter that liberates CO_2 in the soil [31]. The addition of N fertilizer also decreased soil pH compared to control plots due to oxidation of N fertilizer, i.e. transformation of urea to ammonium ion releasing H^+ in soil solution [32]. The decrement in soil pH with fertilizer N might also be attributed to the rapid release of nutrients, which plants could take up. The increased uptake led to less accumulation of exchangeable bases in soil that otherwise could neutralize soil acidity [33]. Irrespective of the mode of application, the decrease in soil pH was more significant with the increasing application rate of FYM, which might be attributed to a higher quantity of addition of FYM, thereby releasing more organic acids. Irrespective of modes and levels of FYM, soil pH was lowered during the growth stages of the crop compared to sowing time due to acids released by plants' roots in addition to SOM decomposition. Irrespective of mode and levels of FYM and fertilizer N, a decrease in soil EC during crop growth was attributed to a change in equilibrium between the concentration of salts and ionic species in the soil solution. The mineralization of FYM during the early stages of crop growth might increase salt concentration in soil solution and re-adsorption of ions on the soil surface, in addition to crop uptake and decreased soil EC during later crop growth stages [34].

Soil organic carbon

Applying FYM for 51 years significantly increased OC content in the soil compared to the initial value due to the direct addition of organic matter in the soil. Irrespective of the mode of application, OC content was accrued with an increasing application rate of FYM due to higher quantity addition. Further, adding fertilizer N significantly increased OC content over control due to higher crop productivity with N addition, leading to higher SOC accumulation [35]. The higher build-up of OC with the integrated application of FYM and fertilizer N as compared to FYM alone may be due to more generation of root biomass, rhizodeposition, left-over residues, and ultimately, incorporation of more organic matter in soil [36]. Irrespective of modes and levels of FYM, OC content in soil was increased from sowing to heading due to the addition of carbon substances through FYM and, thereafter, decreased up to harvesting due to the greater utilization of carbonaceous compounds by microbes and more losses of organic matter as CO_2 [12]. This study also observed a polynomial relationship between SOC and soil properties, with $R^2 = 0.93$ to 0.99 (Fig. 2).

Dissolved organic carbon

During the 51st year of continuous fertilization and cropping under the pearl millet-wheat system, DOC content was significantly increased with FYM application because of abundant food availability that stimulated microbial growth, accelerated organic matter decomposition and increased the DOC content in soil [37, 38]. The lowest DOC content was observed in control plots due to low inherent fertility. The application of fertilizer N improved DOC content in the soil compared to the control, which might be attributed to increased organic matter content due to increased crop growth and more rhizo-depositions. The higher DOC content in integrated plots compared to the sole application of nutrient sources might also be due to the priming effect of fertilizer N to FYM [4, 39]. An increase in DOC content during initial stages may be due to increased mineralization, and a decrease at later stages might be due to oxidation loss of carbon as CO_2 along with utilization by the crop. Another reason for higher content at the initial stages may be the favourable temperature and moisture conditions for microbial proliferation [40]. In this study, a polynomial relationship between SOC and DOC content of the soil was also observed, with $R^2 = 0.98$ (Fig. 2).

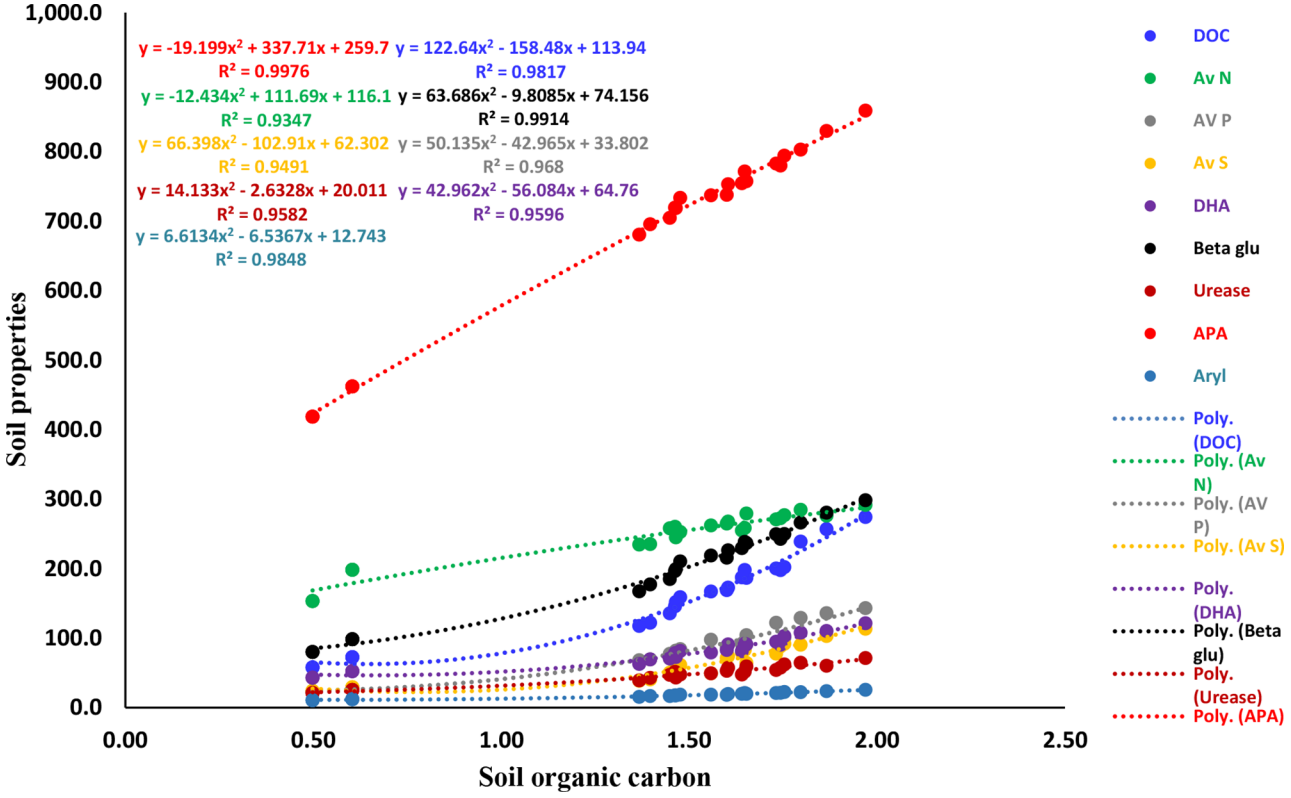


Fig. 2 Relationship between soil organic carbon (SOC) and dissolved OC (DOC), available N, P, S, dehydrogenase activity (DHA), β-Glucosidase (Beta glu), urease, alkaline phosphatase activity (APA) and aryl sulfatase activity (Aryl)of soil

Nutrient availability in soil

Continuous rotation of pearl millet-wheat for 51 years could not sustain the initial level of N in control plots, which might be attributed to N leaching and volatilization losses in addition to plant utilization [8]. Applying FYM improved available N content in the soil because of the direct addition of organic matter and reduced N losses that build up the available N in the soil. Moreover, it might also be attributed to the stimulation of microbial activity, which mineralizes organic matter and releases N in soil solution [5]. The probable reason for the increased availability of N with the INM compared to FYM alone may be the high C: N ratio of FYM, which caused immobilization with the sole application of FYM [41]. Adding N fertilizer and FYM narrowed the C: N ratio and increased mineralization [42]. Another reason might be that excess C supply may lead to variations in the proportions of fungi and bacteria that alter the N cycling in soil [43]. The plots receiving FYM in both seasons recorded the highest available N content due to higher quantity addition and urease activity. The available N content in soil was increased from sowing to heading due to rapid hydrolysis of urea and increased mineralization of FYM [8, 33, 44]. The lower available N content from heading to maturity may be due to plants' uptake and losses of N through biotic and abiotic processes after mineralization [41]. The lower N content during later stages might also be due to reduced N-mineralization because of decreased labile organic matter, microbial activity, and toxic substance accumulation during later stages [44, 45]. A polynomial relationship between urease activity and available N content of soil was also observed ($R^2 = 0.97$; Fig. 3).

The lowest available P content in the control plot may be attributed to the fixation of inherited P in soil in addition to crop removal. A significant build-up of available P with FYM application may be due to mineralization and solubilization processes of P compounds in soil [7]. The organic amino compounds added through FYM also retard the fixation of P in soil and cause desorption of P from mineral compounds [46]. The high C: N ratio of FYM promotes fungal feeding populations, so the availability of P could be improved due to pictorial-fungal associations. Adding FYM over long period also decreased the activity of polyvalent cations such as Ca, Fe, and Al due to chelation, which ultimately reduces P-fixation [47]. The increase in available P content during early growth stages might be due to decomposition of SOM and solubilization of phosphates. The decline in P content after attaining the peak might be due to plants' utilization, and also the reduction in easily degradable organic phosphates [44]. The P availability in soil is also governed OC content of soil and alkaline phosphatase

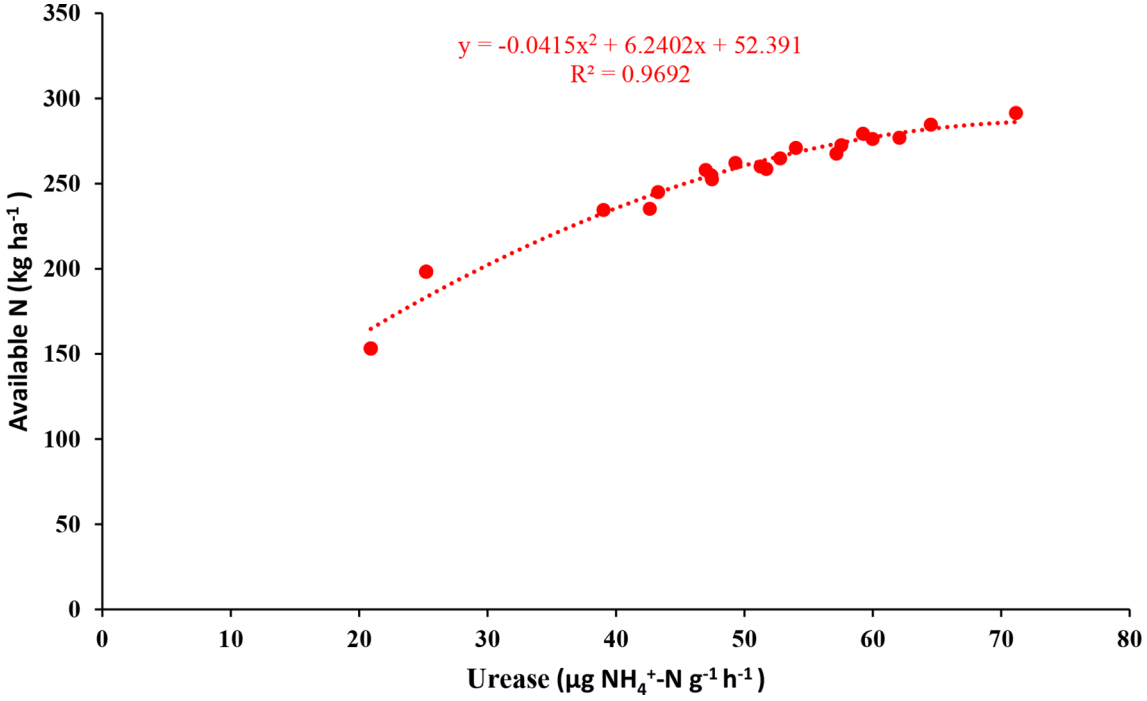


Fig. 3 Relationship between urease activity and available N of soil

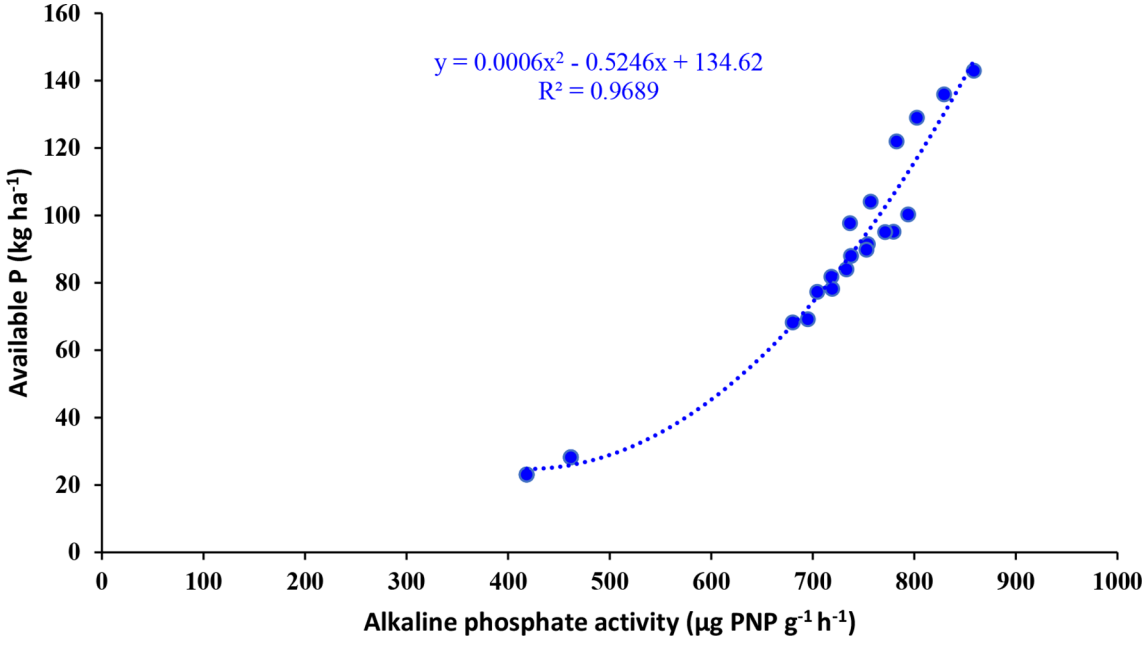


Fig. 4 Relationship between alkaline phosphatase activity and available P of soil

activity (APA) [48]. A polynomial relationship between SOC and available P; APA and available P content of the soil had $R^2 = 0.97$ (Figs. 2 and 4).

The incorporation of FYM for 51 cycles of pearl millet-wheat crop rotation significantly increased microbial biomass and enzyme activities that enhanced the availability of S in soil [4]. Similar to N and P, irrespective of modes of application, the higher application rate of FYM increased available S content in the soil. The lower available S content during the early stages might be due to higher leaching losses of soluble sulfate because of increased moisture availability [8]. The increase in available S content at later stages could be due to favourable temperature that increased microbial activity and mineralization. A polynomial relationship between SOC and available S (Fig. 2; $R^2 = 0.95$); ARA and available S content

(Fig. 5; $R^2 = 0.96$) of the soil was observed in the present study.

Soil enzymes

FYM application significantly increased DHA compared to control or fertilizer N application because FYM supply carbon and energy sources for microbes that improved autochthonous and exogenous microbial diversity ultimately the higher dehydrogenase activity [18, 44]. Dehydrogenase activity increased with the increasing application rate of FYM due to higher substrate addition for microbes. The lowest dehydrogenase activity in control pots might be due to insufficient substrate for microbial growth. Comparatively, higher activity with integrated application of FYM and fertilizer N might be due to increased decomposition of organic matter because of lowered C: N ratio and more availability of nutrients for micro-organisms [39]. The deficiency of available P in soil resulted in decreased enzyme activity at the initial stage [49]. The highest activity at heading stage might be attributed to increased root exudates leading to more microbial activity, which gradually decreases when the crop matures [12]. Moreover, applied FYM gets mineralized rapidly providing sufficient nutrition for the proliferation of microbes in at early stages of crop growth, and scarcity of easily decomposable substrate at the end of crop season gradually decreased the enzyme activity [50]. The lowest dehydrogenase activity after harvest could be due to less moisture availability at maturity [44]. A good correlation between SOC and DHA content of the soil was observed, having $R^2 = 0.96$ (Fig. 2).

Incorporating FYM increased cellulose and hemicellulose degrading β-glucosidases enzyme activity by increasing substrate availability and stimulating the microbial population [51]. The higher β-glucosidase activity with the combined application of FYM and fertilizer N was possibly due to increased nutrient availability and intensified root biomass [8]. Temporal variations in enzyme activity might be attributed to alteration in macro and microclimatic conditions, nutrient availability, and substrate inputs [52, 53]. The gradual decline in β-glucosidase activity might be due to decreasing C substrates with crop progress [54]. The irregularly decreasing trend during pearl millet might also be due to elevated soil moisture that exceeded optimal conditions for microbes. The higher β-glucosidase activity at harvesting stage might be due to an increased decomposition of residues with time and, ultimately, accumulation of SOM [55, 56]. A polynomial relationship between SOC and β-glucosidase activity of soil was observed, having $R^2 = 0.99$ (Fig. 2).

The incorporation of FYM increased substrate availability that increased the microbial population and urease. The addition of fertilizer N also increased the enzyme activity due to the addition of substrate, i.e. urea. The increased enzyme activity with increased FYM application rate may be attributed to enhanced microorganisms' populations because of more availability of carbon-releasing extra-cellular enzymes [57]. However, the higher application rate of FYM may also decrease the enzyme activity due to inhibitory effect of the end products [58]. The highest urease activity with FYM application in both seasons was due to higher substrate addition

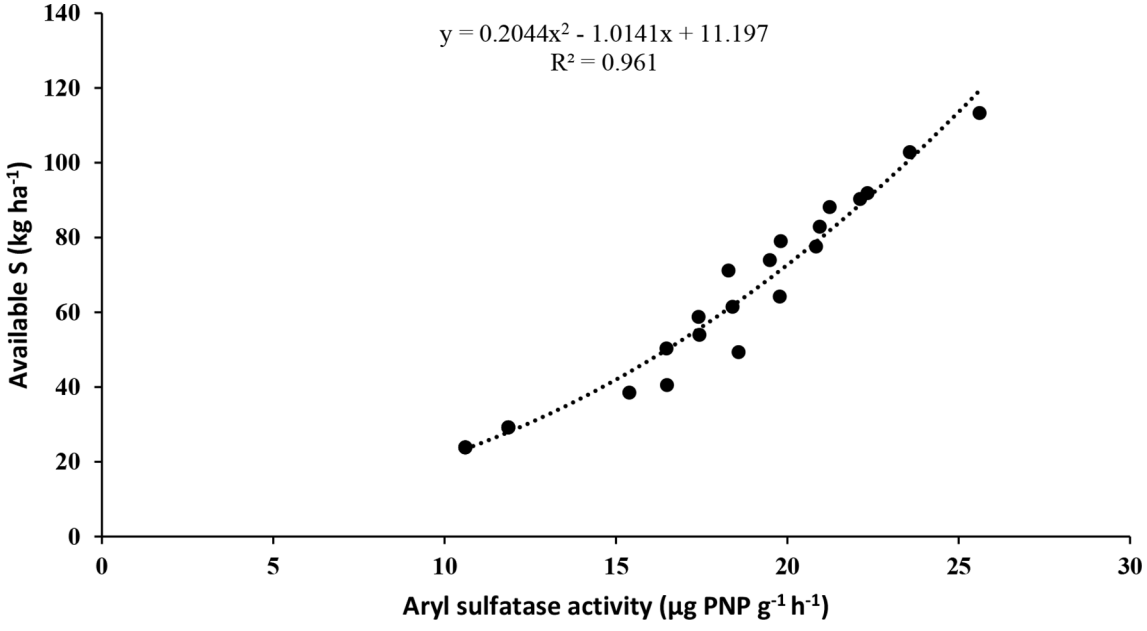


Fig. 5 Relationship between aryl sulfatase activity and available S of soil

and C availability [59]. Variations at growth stages indicated that temperature and moisture conditions influenced enzyme activities by affecting microbial growth and substrate availability. The enhanced urease activity at tillering stage might be attributed to higher N content due to faster decomposition of SOM and release of ammonium [60]. The lower enzyme activity at harvesting stage was due to low temperature, lower microbial biomass, and decreasing SOC content [44]. In this study, a polynomial relationship between SOC and urease activity of soil having $R^2 = 0.96$ was also observed (Fig. 2).

A significant improvement of APA in manure treated soil could be related to enhanced microbial activity and diversity of phosphate-solubilizing bacteria [61]. The addition of FYM provides sources of C, N, and P for soil microorganisms which might stimulate the alkaline phosphatase activity [58]. Alkaline phosphatase activity was higher with conjunctive use of FYM and fertilizer N because of faster decomposition of FYM in presence of fertilizer N that resulted in more release of organically bound P. The balanced fertilization also promotes higher microbial population and more secretion of enzymes [39, 44]. The higher phosphatase activity from sowing to heading might be attributed to sufficient temperature and moisture availability in the rainy summer. The rainfall during summer season often leads to nutrient loss and low available P content, which enhances phosphatase activity and fulfils the increasing P demand for plant and microbe growth. In this study, a polynomial relationship between SOC and APA activity of soil was also observed, with $R^2 = 0.99$ (Fig. 2).

Like other enzymes, arylsulfatase activity is also associated with the application rate of organic manures, and increased activity with FYM application is due to a favourable environment for C-degrading microorganisms and enhanced substrate [51]. With FYM application, elevated activity indicates the differences in microbial community and higher fungal biomass. The higher enzyme activity with combined use of FYM and fertilizer N than FYM alone could be due to enhanced organic matter input and C, N, P, and S availability [8, 62]. Variations at different stages indicated that enzyme activity is influenced by temperature and moisture conditions [40]. Significantly higher enzyme activity was noticed in the wet season than in dry season. The higher activity at tillering might also be ascribed to an increase in root exudates that leads to more intense microbial growth, which gradually declines with the advancement of crop growth [12]. This study also reported a polynomial relationship between SOC and ARA activity of soil, with $R^2 = 0.98$ (Fig. 2). The polynomial relationships between SOC and enzyme activity are essential for understanding soil dynamics, fertility, and ecosystem health [63]. They highlight the interconnectedness of biological processes within soils and their broader environmental implications.

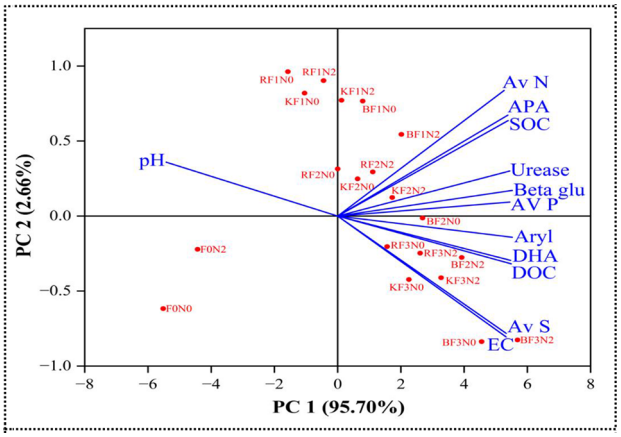


Fig. 6 Plot of principal component analysis (PCA) on the soil properties including soil pH, electrical conductivity (EC), soil organic carbon (SOC), dissolved OC (DOC), available N, P, S, dehyrdogenase activity (DHA), β-Glucosidase (Beta glu), urease, alkaline phosphatase activity (APA) and aryl sulfatase activity (Aryl)in soils (pooled data of five growth stages of pearl millet) under pearl millet-wheat cropping system in North-western, India. PC = principal component

Principal component analysis (PCA)

The principal component analysis (PCA) separated the soil variables and treatments in orthogonal space (Fig. 6). The PC1 and PC2 explained 98.36% variability in the data set (95.7% and 2.66% of total variability, respectively). The PC1 and PC2 had eigen values of 11.48 and 0.32, respectively. Each soil variable had a corresponding factor loading or eigenvector weight value and contribution to variability (Supplementary Table S4). The β-glucosidase and ARA both had the highest loading value (0.294), followed by DOC (0.292) and DHA (0.291) in PC1 and also correlated with other high loading indicators. In the second PC (PC2), available N was the most highly loaded indicator (loading value = 0.462) followed by EC (-0.444) and available S (-0.431) in PC2. The percentage of total variance estimated based on the weight of each PC, the weighted factor for two distinct minimum data sets (MDS) followed PC1 (0.97) > PC2 (0.03). High levels of β-glucosidase activity indicated a robust microbial population capable of breaking down organic materials, essential for maintaining soil health and productivity. DHA play a key role in the respiration processes of microorganisms by transferring hydrogen from organic substrates to electron acceptors. High DHA is often associated with increased microbial metabolic rates, indicating a thriving microbial community contributing to decomposition and nutrient cycling of organic materials. Elevated levels suggest enhanced soil structure and fertility, while low levels may indicate poor soil conditions [64]. DOC represents primary energy source for soil microbes, influencing their

growth and activities. Thus, high levels of DOC indicate healthy soil ecosystems and reflect active organic matter turnover and nutrient cycling processes [65]. In PCA, PC1 often captures the most variance among the data set, highlighting these variables' importance in distinguishing between different soil conditions or treatments [64]. The high loadings on PC1 suggest that β-glucosidase, DHA and DOC are interrelated factors that collectively influence the biological processes within the soil ecosystem. The β-glucosidase and DHA activities alongside DOC and their interactions facilitate essential processes such as organic matter decomposition, nutrient cycling, and overall ecosystem functioning, making them critical indicators for assessing soil health.

Conclusions

Long-term application of FYM and chemical N significantly influenced the soil properties during the 51st cycle of the pearl millet-wheat system. Application of 15 Mg FYM ha^{-1} showed the highest available N, P, and S content, DHA, urease, β-Glucosidase, APA, and ARA activity significantly at different growth stages over FYM_{10}. Chemical N application significantly accrued soil OC and increased nutrient flush to the equilibrium soil solution. A synergistic favour and significant enrichment in SOC, available nutrients, and enzyme activity were also recorded with treatment in both seasons x FYM_{15}. Interaction of FYM_{15} x N_{120} was also significantly superior for maintaining the highest level of DOC, available N, APA, and urease at different growth stages. Under different nutrient management practices, Urease, APA, and ARA were strongly related to their respective nutrients ($R^2 > 0.96$). The PC1 and PC2 explained the 98.36% variability in the data set (95.7% and 2.66% of total variability, respectively). The β-glucosidase and ARA both had the highest loading value (0.294), followed by DOC (0.292) and DHA (0.291) in PC1 and also correlated with other high loading indicators. Upon the examination of data during the 51st cycle of pearl millet, the PCA biplot revealed that the application of 15 Mg FYM ha^{-1} coupled with 120 kg N ha^{-1} during both seasons was separated and found best for soil fertility throughout the growing period of pearl millet under pearl millet-wheat sequence in North-West Haryana, India.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12870-025-06128-2.

Supplementary Material 1

Acknowledgements
Authors are thankful to the Researchers Supporting Project number (RSP2025R347), King Saud University, Riyadh, Saudi Arabia.

Author contributions
S S and D P: Writing-original draft, Investigation, Data curation. S S, D R, R K G, MHS and SK: Writing-original draft, Supervision, Methodology, Conceptualization D P; P K Y and R S: Writing-review & editing, Visualization, Methodology, Data curation.S A, SK and M HS: Writi ng-review & editing, Software. SA, SK: Critically revised and funding acquisition. All authors have read and agreed to the published version of the manuscript. SS, DP & DR contributed equally.

Funding
The authors appreciate the Researchers Supporting Project number (RSP2025R347), King Saud University, Riyadh, Saudi Arabia.

Data availability
Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate
This article does not contain any studies with human or animal subjects. The current experimental research and field study including the collection of plant material, is complying with relevant institutional, national, and international guidelines and legislation and used for research and development.

Consent for publication
Not applicable.

Clinical trial number
NA.

Competing interests
The authors declare no competing interests.

Received: 19 October 2024 / Accepted: 20 January 2025
Published online: 27 January 2025

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